

Theory of Computing

Midterm Exam - 60 mins

2024/2025

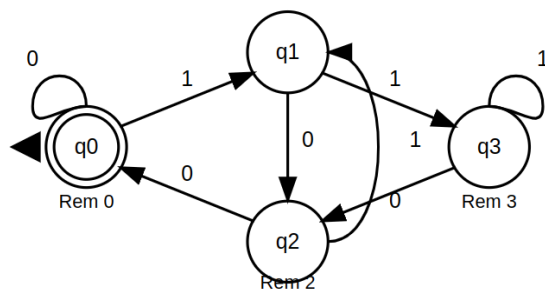
2ND YEAR - ENSIA

EXERCISES : Answer all questions.

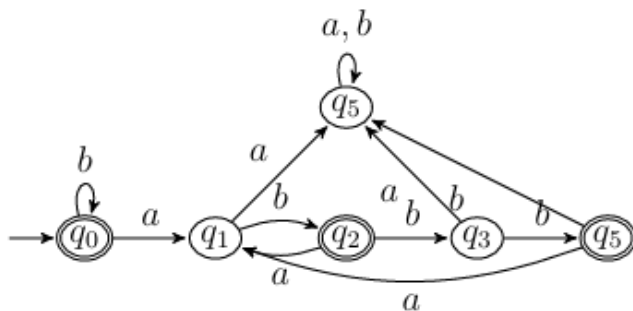
Exercise 1 (9 points) [DFA & NFA] :

1. [7 points] For each of the following problems construct a deterministic finite automaton which describes or recognizes the language given. The underlying alphabet is $\{0,1\}$.

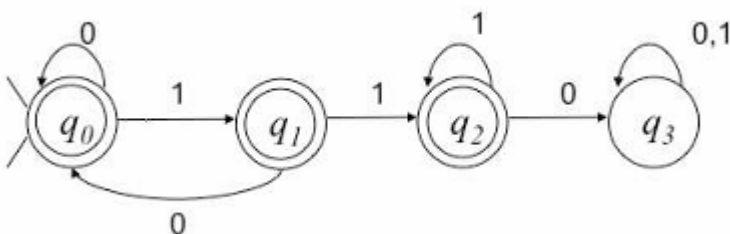
1. [1 points] $\{ w \mid \text{decimal value of } w \text{ divisible by } 4 \}$ (Hint : ends with two zeros)



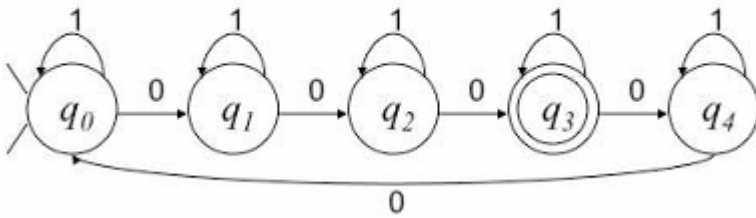
2. [1.5 points] $\{ w \mid \text{in } w \text{ each } 0 \text{ followed by exactly one "1" or 3 "1"s } \}$



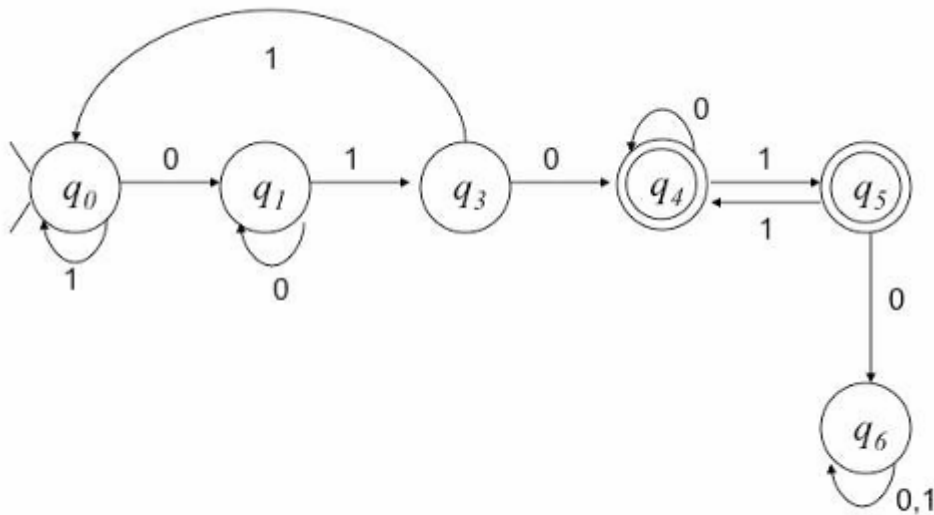
3. [1.5 points] $\{ w \mid w \text{ does not contain the substring } 110. \}$ [From Optional]



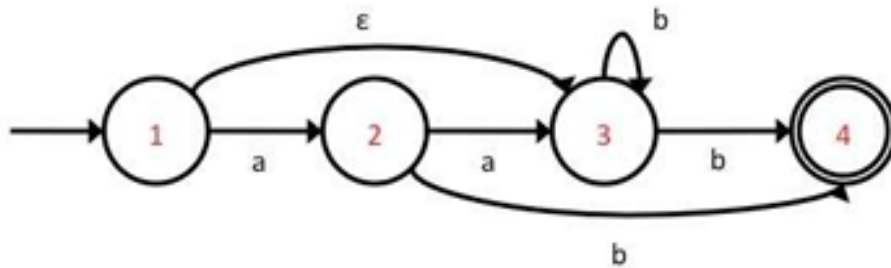
4. [1.5 points] $\{ w \mid w \text{ has } n \text{ occurrences of } 0\text{'s where } n \bmod 5 \text{ is } 3. \}$



5. [1.5 points] $\{ w \mid w \text{ has exactly one occurrence of the substring } 010. \}$



2. [2 point] Convert the following NFA to a DFA:



Exercise 2 (6 points) [Regular Expressions] :

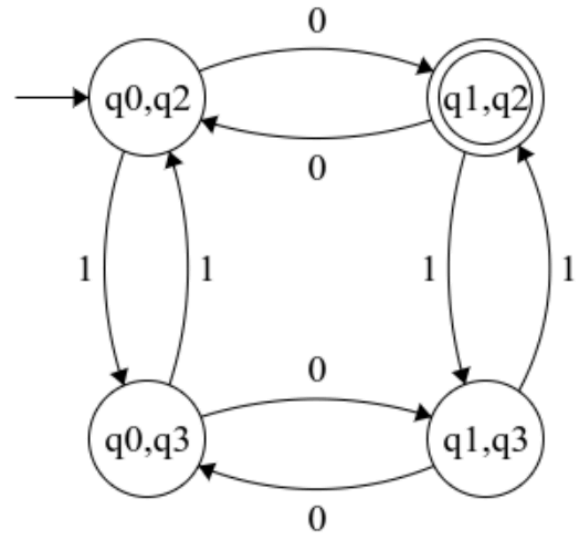
1. [3 points] Write regular expressions for the following regular languages:

1. [1 points] The language which does not contain the 10 as a substring. (From the lecture)
 0^*1^*
2. [1.5 points] The language $\{w \in \Sigma^* \mid w \text{ contains exactly one double letter}\}$. For example, baaba has exactly one double letter, but baaaba has two double letters.
 $(\epsilon \mid b)(ab)^*aa(ba)^*(\epsilon \mid b) \mid (\epsilon \mid a)(ba)^*bb(ab)^*(\epsilon \mid a)$

3. [1.5 points] The language $\{w \in \Sigma^* \mid w \text{ does not end in a double letter}\}$.

$\epsilon \mid a \mid b \mid (a|b)^*(ab|ba)$

2. [2 points] Convert the following DFA/NFA to a RegEx :



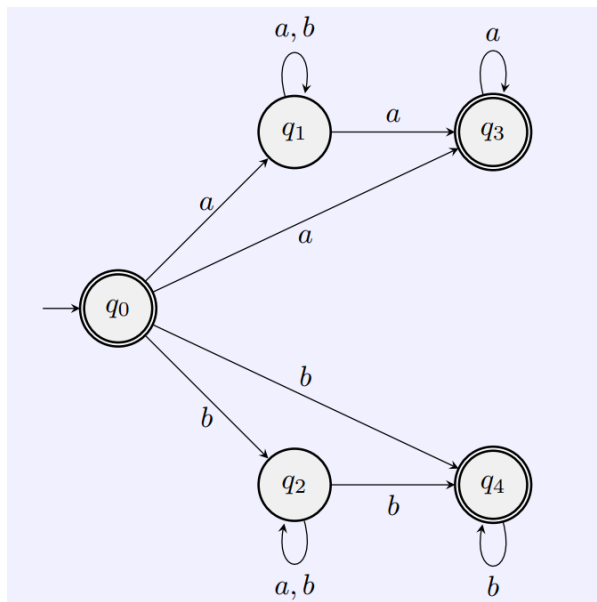
Exercise 3 (4.5 points) [Regular Languages] :

1. Justify whether the following language is regular or not regular. (Either by providing DFA/NFA or use the pumping lemma:

a. [1.5 points] $L = \{w \mid \text{the number of substring "ab" equals the number of substring "ba" in the word } w\}$ over the alphabet $\{a,b\}$

Regular :

$$\epsilon \cup a \cup b \cup (a(a \cup b)^*a) \cup (b(a \cup b)^*b)$$



b. [1.5 points] $L = \{a^n b^m \mid n, m \geq 0, n > m\}$ over the alphabet $\{a,b\}$ [From Optional]

Not regular,, Pumping down.. $s = xyz = a^{p+1}b^p$ for the case where $i=0$, $s' = xy^0z$ should be in the language, but $|y| > 0$, therefore, by pumping down, the number of a

would be equal or less than the number of b. Which is not in the language...

- c. [1.5 points] $L = \{w \mid w = w^R \text{ and } |w| \text{ is odd}\}$, meaning that w is a palindrome of odd length } over the alphabet $\{a\}$
Regular : words are constructed only using a and therefore we can write the regex : $(aa)^*a$

Exercise 4 (6 points) [CFG] :

Produce the Context-Free Grammar for the following languages :

1. [3 point] The **complement** of the language $\{a^n b^n \mid n \geq 0\}$ over the alphabet $\{a, b\}$ [From optional]

$$\begin{aligned} S &\rightarrow S_1 \mid S_2 \\ S_1 &\rightarrow aS_1b \mid T \mid U \\ S_2 &\rightarrow RbaR \\ T &\rightarrow aT \mid a \\ U &\rightarrow Ub \mid b \\ R &\rightarrow RR \mid a \mid b \mid \epsilon \end{aligned}$$

2. [3 point] The language $L = \{x\#y \mid |x| \neq |y|\}$ over the alphabet $\{a, b, \#\}$. (Examples of words in the language : $aa\#bbb$, $aaaa\#b$..
Words not in the language : $a\#b$ }

$$\begin{aligned} S &\rightarrow XSX \mid XL \mid RX \\ L &\rightarrow \# \mid XL \\ R &\rightarrow \# \mid RX \\ X &\rightarrow 0 \mid 1 \end{aligned}$$

OPTIONAL EXERCISES (Bonus Marks):

1. [1.5 point] Construct the Context-Free Grammar for the language where the number of $b + 2 =$ the number of a . The number of a is more than b but with strictly only two letters.
2. [1.5 point] $L = \{0^k \mid k \text{ is a prime number}\}$. Provided that the set of prime numbers is infinite. (no need to prove this).